Experimental Methodology: Investigating Expert Pre-training Effects on MoE Domain Specialization

1 Overall Goal

Investigate whether narrow task-specific pre-training of experts before general MoE training increases the weak domain specialization observed by Fan et al. for sequence-level routing.

2 Four-Step Experimental Process

2.1 Step 1: Train Baseline Model (Replicating Fan et al.)

Architecture: GPT2-mini based transformer with LoRA extension

Configuration:

- 3 transformer layers with 4 experts per layer (scaled down from Fan et al.'s 12 layers)
- Sequence-level Top-2 routing with double softmax:

$$p_i(x) = \frac{e^{h_i(x)}}{\sum_j e^{h_j(x)}}$$

$$y = \sum_{i \in \tau} \frac{e^{p_i(x)}}{\sum_{j \in \tau} e^{p_j(x)}} E_i(x)$$

  where τ is the set of top-2 indices
- Load balancing loss ($\lambda = 0.01$)

Training data: Subset of OpenWebText (due to 16GB VRAM constraint vs. Fan et al.'s 40GB)

Purpose: Establish baseline routing behavior without expert pre-training

2.2 Step 2: Pre-train Individual Experts on Specialized Tasks

Data source: MMLU auxiliary training set (domain-specific questions/answers)

Method: Train individual experts separately on specific subject domains from MMLU

Purpose: Give each expert narrow task-specific knowledge before MoE integration

Key innovation: This is the novel contribution—experts start with domain specialization rather than learning it emergently

2.3 Step 3: Combined MoE Training on OpenWebText

Architecture: Insert pre-trained experts into full MoE structure

Training data: OpenWebText (general dataset)

Configuration: Same as baseline (sequence-level Top-2 routing, 4 experts per layer)

Purpose: Allow the pre-trained experts to participate in general language modeling while potentially retaining their specialization

2.4 Step 4: Evaluate Routing Behavior on MMLU Tasks

Test data: MMLU test set with subject-specific questions (6 categories used by Fan et al.)

Measurement: Record which experts are activated for each subject domain

Visualization: Create routing heatmaps showing expert activation patterns by subject (similar to Figure 2 in Fan et al.)

Analysis: Compare routing patterns between:

- Baseline model (no pre-training)
- Pre-trained expert model

3 Expected Outcome

If pre-training increases specialization, we should observe **stronger differentiation** in expert activation patterns across subjects compared to the "weak specialization" observed in the baseline. The routing behavior should show clearer subject-specific expert preferences in the pre-trained model.

4 Key Comparison Metric

The degree of routing differentiation across MMLU subjects—quantified by differences in expert activation frequencies (the "staple heights" in routing visualizations) between subjects.

4.1 Operational Definition

For non-specialization, we would expect equal use of each expert across subjects. Specialization manifests as different activation patterns (staple heights) for different subjects, indicating that the MoE routes differently based on subject matter.